

Sineth Induwara Perera Wickramaratna

Software Engineering · AI/ML · Data Science | CSE Undergraduate, University of Moratuwa
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SUMMARY

Computer Science and Engineering undergraduate with hands-on work across software engineering, AI/ML and data science. **Software:** Java/Spring Boot and Node.js/TypeScript microservices, REST and event-driven (Kafka) architectures, React front ends, PostgreSQL, MongoDB and Redis, shipped with Docker, Kubernetes and CI/CD. **AI/ML:** LLM fine-tuning (LoRA/QLoRA), a LangGraph multi-agent RAG system, and PyTorch deep learning tracked in Weights & Biases. **Data science:** multi-source data cleaning with Pandas and NumPy, EDA, scikit-learn modelling, and rigorous evaluation with hypothesis tests, cross-validation and multi-seed experiments. Works in Agile teams through Git pull-request workflows, with testing and CI on every project.

EDUCATION

University of Moratuwa, Sri Lanka Mar 2024 – Present
B.Sc. Engineering (Hons) in Computer Science and Engineering | CGPA: 3.38 / 4.00
Relevant coursework: Object-Oriented Programming, Data Structures & Algorithms, Database Systems, Software Architecture, Machine Learning, Deep Neural Networks, Introduction to Data Science, Data Science & Engineering Project

G/Dharmasoka College 2020 – 2023
G.C.E. A/L, Physical Science: Combined Maths (A), Physics (A), Chemistry (A) | Z-Score 2.5767 (District 25, Island 160)

TECHNICAL SKILLS

Languages	Python, Java, TypeScript, JavaScript, SQL, Dart, C++ , C
Back-End	Spring Boot, Node.js/Express.js, FastAPI; REST APIs, microservices, event-driven architecture (Kafka), JWT auth, RBAC
Front-End & Mobile	React, Next.js, Tailwind CSS, Vite, HTML, CSS; Flutter
Databases	PostgreSQL, SQL schema design, MongoDB, Firebase/Firestore, Neo4j, InfluxDB; Redis
Cloud & DevOps	Docker, Docker Compose, Kubernetes, GitHub Actions CI/CD, Nginx, AWS EC2; OpenTelemetry, Prometheus, Grafana
Data Science	Pandas, NumPy, SciPy; data cleaning, imputation, feature engineering, EDA; Matplotlib, Seaborn, Plotly; Jupyter
Statistics	Hypothesis testing (McNemar, Wilcoxon), correlation analysis, cross-validation, calibration
Machine Learning	scikit-learn, XGBoost, CatBoost; Random Forest, SVM, Logistic Regression; GridSearchCV
Deep Learning	PyTorch, Hugging Face Transformers, PEFT (LoRA/QLoRA), timm, Grad-CAM
LLM / GenAI	Agentic AI (LangGraph), RAG, LangChain, ChromaDB, Sentence-Transformers, spaCy, Gemini API, prompt engineering
MLOps & Testing	Weights & Biases, YAML configs; pytest, JMeter, Postman, CI-gated test suites; Claude Code

SELECTED PROJECTS

ORBIT — Intelligent Distributed System Digital Twin Ongoing
Platform for modelling, monitoring and simulating distributed microservice architectures across domains

- Developing a digital twin that builds dynamic dependency graphs from service interactions and telemetry, enabling real-time health monitoring and what-if failure impact analysis without touching the live system.
- Designing an event-driven microservice architecture on Apache Kafka with Spring Boot and FastAPI services, PostgreSQL for state, Neo4j for the dependency graph and Redis caching.
- Implementing controlled fault injection, OpenTelemetry distributed tracing and Prometheus/Grafana metrics, with ML-based telemetry analysis for anomaly detection and failure prediction; deployed with Docker on Kubernetes.

Stack: Java, Spring Boot, FastAPI, Kafka, PostgreSQL, Neo4j, Redis, Docker, Kubernetes, Next.js, OpenTelemetry, Prometheus

Clario — Multi-Agent Customer Support Triage System 2026
Group project | [GitHub](#)

- Built an agentic AI platform that classifies, prioritises, routes, drafts replies for and escalates support tickets, using a LangGraph workflow: PII masking (regex + spaCy NER) → classification → routing to Technical/Billing agents → RAG drafting over ChromaDB → judge-model validation → human escalation.
- Fine-tuned Gemma-3 1B and Llama-3.2 3B with LoRA/QLoRA (4-bit NF4) on teacher-distilled labels, benchmarked against scikit-learn and BERT-style baselines on F1, urgent-ticket recall, latency and VRAM.
- Tuned the pipeline against human-labelled data: escalation F1 rose from 0.588 to 0.727, HR-ticket recall from 37% to 79%, and unnecessary escalations fell from 32 to 7.

- Added a semantic cache cutting a 500-ticket duplicate cluster from 75 to 2.4 min (−96.8%).
- Deployed as Dockerised microservices (Spring Boot API gateway, FastAPI ML services, Redis queue, Next.js + Supabase PostgreSQL front end), tested with pytest, JMeter load tests and GitHub Actions CI.

Stack: Python, PyTorch, Hugging Face, PEFT, scikit-learn, Pandas, LangGraph, LangChain, ChromaDB, Gemini API, Spring Boot, FastAPI, Redis, Docker, Next.js, PostgreSQL

Trustworthy Deep Learning for Chest X-ray Disease Detection

2026

Research project, Deep Neural Networks (CS3631) | Co-author of the research paper | [GitHub](#)

- Contributed to benchmarking ResNet50, DenseNet121, EfficientNet-B0 and ViT-Base on ~21K chest X-rays, ranking them on accuracy, calibration, explanation faithfulness, efficiency and OOD robustness.
- Helped develop a lung-region attention module raising in-lung Grad-CAM evidence on DenseNet121 by $+7.05 \pm 0.86$ pp across 3 seeds (Wilcoxon $p < 10^{-140}$) with no detectable accuracy cost; led a 7-arm ResNet50 ablation reaching 92.0% accuracy and 0.988 macro-AUC.
- Built a reproducible codebase with 160 pytest unit tests in GitHub Actions CI and Weights & Biases run tracking.

Stack: Python, PyTorch, timm, scikit-learn, SciPy, NumPy, Pandas, Grad-CAM, Weights & Biases, pytest

Identifying and Predicting Malnutrition Hotspots

2025

ML research project, Introduction to Data Science | [GitHub](#)

- Merged 12+ FAOSTAT, World Bank, UNDP and WHO indicators into a country-year panel (136 countries, 2000–2022) with Pandas; cleaned it with linear interpolation, median/mode imputation and categorical encoding.
- Designed a composite dual-burden Malnutrition Index (normalised stunting + overweight) as the target, with class thresholds based on WHO/UNICEF guidance.
- EDA (correlations, boxplots, choropleth maps) linked stunting to HDI ($r = -0.84$) and water access ($r = -0.78$).
- Compared 5 classifiers with GridSearchCV, cross-validation and class weighting; Random Forest was best (93.8% accuracy, 0.99 ROC-AUC), beating linear baselines by 30+ points; written up as an ACM paper.

Stack: Python, Pandas, NumPy, scikit-learn, XGBoost, CatBoost, Plotly, Seaborn

Academent — AI Study Companion Platform

2026

Solo project | [GitHub](#)

- Shipped a seven-module full-stack web app (planner, notes, flashcards, generated quizzes, AI tutor, reminders, analytics) on React/Vite with Firebase Authentication and Google GenAI integration.
- Built the Node.js/Express REST API with Firebase Admin token verification on every protected route, per-user rate limiting and structured error handling, plus an admin dashboard; authored the IEEE 830 SRS.

Stack: React, Vite, Node.js, Express.js, Firebase, Google GenAI

VeriText AI — AI Text Detection & Document Analysis Platform

2026

Solo project | [GitHub](#) | [Hugging Face](#)

- Fine-tuned all-MiniLM-L6-v2 in PyTorch to detect AI-generated text, published it on Hugging Face, and served it via a FastAPI REST service with a React + Tailwind CSS front end, containerised with Docker Compose behind Nginx.

Cattle.io — Smart Livestock Monitoring Platform

2026

Group project — mobile and backend | [GitHub](#)

- Designed the telemetry ingestion path into a Node.js/Express service (MongoDB for transactional records, InfluxDB for time-series) behind a Flutter app; wrote hardware simulators to regression-test alert flows.

Amica — Women's Safety Companion

2026 – Present

Group project, ongoing | [GitHub](#)

- Building the emergency-response backend (SOS dispatch, journey-timer check-ins, vehicle risk lookups) as TypeScript Cloud Functions with Firestore security rules, reviewed through GitHub pull requests.

HONOURS & AWARDS

- **Champion**, RoboRoarZ Sri Lanka International Robotics Competition, IEEE Student Branch, UoM (2026)
- **Finalist**, SLIIT RoboFest 2025 | **Finalist**, UoJCoders v4.0 (2025) | **Semi-Finalist**, SPARK Challenge 2026, UoM

LEADERSHIP & ACTIVITIES

Rotaract Club of University of Moratuwa

Aug 2024 – Present

PR Avenue Senior Director (2026–present) · PR Avenue Director (2025–26) · Assistant Treasurer (2024–25)

IEEE Student Branch, RAS Chapter & CSE Student Society, University of Moratuwa

Jan 2025 – Present

Design Co-Lead, MoraForesight 4.0 (2026–) · PR Visual Media Director, CSE Society (2026–) · Design Co-Lead, SLIoT (2025–26)

REFERENCES

Prof. Dulani Meedeniya

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Dr. Kutila Gunasekara

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